IC101

Introductory Securities & Markets

Lecture 2 · Balance sheets, funding & financial intermediaries

Professor George Alexandridis · ICMA Centre, Henley Business School

From last week to this week

We had the physics of value. Now the anatomy of the machine.

Last week: the toolkit

Price = future cash, adjusted for TIME and RISK.

The three pillars: time value, risk, asset pricing.

You can now value a bond, a share, a project.

But we never asked HOW the money actually moves from savers to the companies that use it.

This week: the machine

The money moves through balance sheets, and every institution in finance is really just a balance sheet.

Remember Silicon Valley Bank and Northern Rock? A bank IS a balance sheet. When the balance sheet broke, the bank died.

Today you learn to read one, and to X-ray any company with it.

Bottom line: **The one sentence that unlocks it all: every asset is somebody else's liability.**

Today: how money really moves

One story, not five separate topics: each step leads straight into the next.

Read a balance sheet — The one map that fits every company. We will read Apple's.

The great choice: debt vs equity — The right-hand side of that map: how the firm is funded.

Who stands between — The intermediaries that carry savers' money to the firm.

Where they meet: markets — Where a firm's new shares and bonds are sold, then traded.

The highlights — **The handful of things your exam is built on.**

Step 1: read a balance sheet

Every company has one. Here is the world's most famous: Apple.

The one equation

Assets = Liabilities + Equity

Left: what the firm **OWNS**.

Right: who **FUNDED** it, lenders (liabilities or what it owes) and owners (equity).

The two sides must always balance. That is the whole idea.

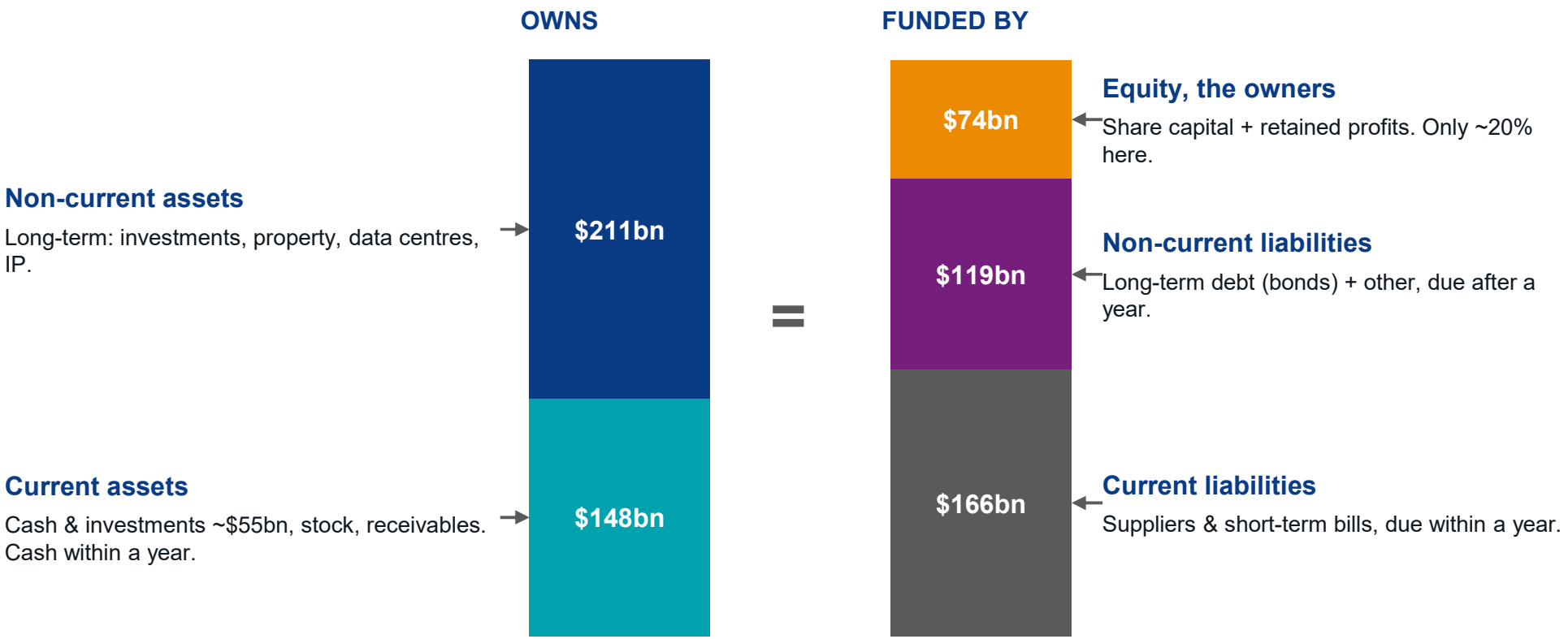
Apple Inc., 2025 balance sheet ($bn)			
Non-current assets (investments, PP&E, IP)	**211**	Equity (the owners' stake)	**74**
Current assets (cash & investments $55bn, stock)	**148**	Liabilities (owed to others)	**285**
Total assets	**359**	**Total funding**	**359**

A company worth over $3 trillion has just ~$74bn of equity on its books, about 20%. Hold that, it's a clue.

Source: Apple Inc. Form 10-K, FY2025 (year ended 27 Sep 2025).

The same balance sheet, opened up

What is actually inside Apple's $359bn, and who paid for it



Bottom line: **Both bars are the same height: assets ALWAYS equal how they were funded. And notice, a $3tn company runs on just ~20% owners' equity.**

Apple Inc. Form 10-K, FY2025; grouped into the standard current / non-current categories, rounded.

For reference: Apple's balance sheet, in full

Every line, in plain words (FY2025, $bn)

WHAT APPLE OWNS assets, $359bn

Current assets · within a year		**$148bn**
Cash & equivalents	**$36bn**	Cash & instant-access money
Marketable securities	**$19bn**	Short-term investments
Accounts receivable	**$40bn**	Money customers owe Apple
Vendor receivables	**$33bn**	Money partners owe Apple
Inventories	**$6bn**	Unsold stock & parts
Other current assets	**$15bn**	Prepayments & odds and ends
Non-current assets · for years		**$211bn**
Long-term investments	**$78bn**	Bonds held for years
Property & equipment	**$50bn**	Buildings, data centres, kit
Other non-current	**$84bn**	IP, deferred tax, investments

HOW IT IS FUNDED claims, $359bn

Current liabilities · within a year		**$166bn**
Accounts payable	**$70bn**	Bills owed to suppliers
Other current liab.	**$66bn**	Accrued costs & taxes
Deferred revenue	**$9bn**	Paid for, not yet delivered
Commercial paper	**$8bn**	Very short-term IOUs
Term debt (current)	**$12bn**	Bonds due within a year
Non-current liabilities · later		**$120bn**
Term debt (long-term)	**$78bn**	Bonds repaid after a year
Other non-current liab.	**$42bn**	Long-term taxes & dues
Equity · the owners' stake		**$74bn**
Share capital	**$94bn**	Raised by issuing shares
Retained earnings	**-$14bn**	Profits kept, minus buybacks
Other reserves	**-$6bn**	Currency & value tweaks

Source: Apple Inc. Form 10-K, fiscal year ended 27 September 2025 ($ rounded).

The right-hand side is a dilemma: debt or equity

Two ways to fund anything, two completely different contracts



DEBT (a loan)

A promise to repay, with interest.

Fixed claim: lenders get paid a set amount, first.

Must be serviced, miss a payment and you can be forced under.

No ownership, no vote, no upside.

e.g. a bank loan, or a bond.



EQUITY (ownership)

A share of the company itself.

Residual claim: owners get what's left, last.

No obligation to pay anything, dividends are optional.

Full ownership, votes, and all the upside.

e.g. ordinary shares in Apple.

Bottom line: Debt is a fixed promise you must keep. Equity is a share of the adventure. Every firm has to chose a mix.

The secret: balance sheets are wired together

Follow one pound of your savings through the system



Your asset is the bank's liability. The bank's asset is the company's liability. The whole system is balance sheets, linked.

And returns, interest and profit, flow back down the chain the other way, exactly like last week's savers-to-borrowers loop.

Quick poll, grab your phones

A profitable firm needs £10m to grow. How should it raise it?

- **A** Borrow it: a bank loan or a bond
- **B** Sell new shares: raise equity
- **C** A mix of both
- **D** Use its own retained profits

Join at vevox.app

Capital structure: the CFO's biggest decision

How much debt should a company take on?

It is a real, human dilemma

A founder on Dragons' Den faces it live: take a loan, or give away equity for the cash?

Give away equity and you dilute your ownership forever.

Take on debt and you owe fixed payments, whatever happens.

Every company, from a start-up to Tesco, is choosing a point on this dial.

The trade-off in one line

More debt can make good years GREAT for owners…

…and bad years FATAL.

That amplifier has a name: leverage. Let's put real numbers on it.

Bottom line: **capital structure is simply the mix of debt and equity a firm uses to fund itself. There is no free lunch, only a trade-off.**

Quick poll, grab your phones

A company takes on more debt. What happens to the owners' returns?

- **A** They always go up
- **B** They always go down
- **C** Bigger in good years, and worse in bad years
- **D** Nothing changes

Join at vevox.app

Leverage: the amplifier, in numbers

Two identical firms with £100m of assets, earning in a good and a bad year

Firm A: all equity

£100m equity, no debt.

Firm B: half debt

£50m equity + £50m debt at 5% (£2.5m interest).

Good year (profit £10m):

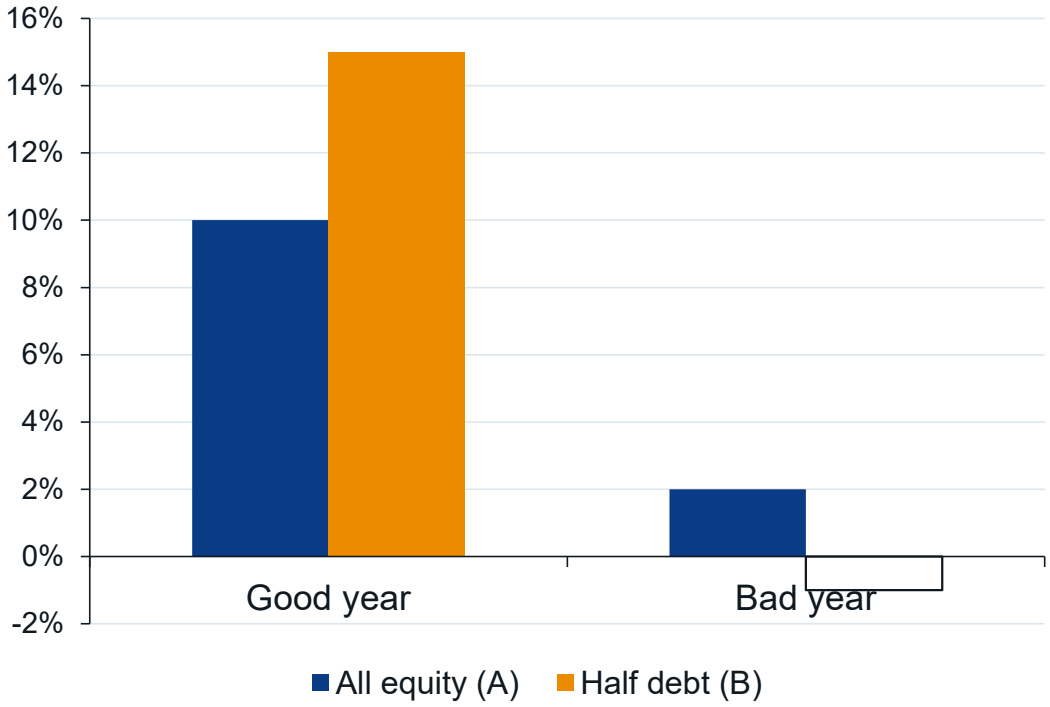
A: 10 ÷ 100 = 10%. B: (10 - 2.5) ÷ 50 = 15%.

Bad year (profit £2m):

A: 2 ÷ 100 = 2%. B: (2 - 2.5) ÷ 50 = -1%, a loss.

Why ÷ 50, not 100?

Owners funded only £50m; lenders funded the other £50m and take only their fixed £2.5m interest. Whatever is left is the owners' return on their £50m.



ROE = profit ÷ owners' equity. Same firm, same profits, debt widens the outcome both ways.

CLICK **Explore leverage with AI** ↗

Two kinds of risk

Where does a company's risk actually come from?



Business risk

The risk in the business itself: will people buy the sausage rolls? Will costs jump?

Comes with the industry. Largely given, you can only manage it so far.



Financial risk

The EXTRA risk you add by financing with debt: those fixed payments you must meet.

Entirely your choice. It is the dial you set on the last slide.

Bottom line: You largely inherit your business risk. You deliberately choose your financial risk.

So how much debt is optimal? Modigliani-Miller

First the base case, then the two frictions that create a sweet spot.

Franco Modigliani & Merton Miller, 1958. Both later won the Nobel Prize in economics (Modigliani 1985, Miller 1990).

Start with M&M I: the base case

In a perfect world, no taxes, no bankruptcy, how you split debt and equity does not change a firm's value. Same pie, different slices.

Then the real world adds two frictions:

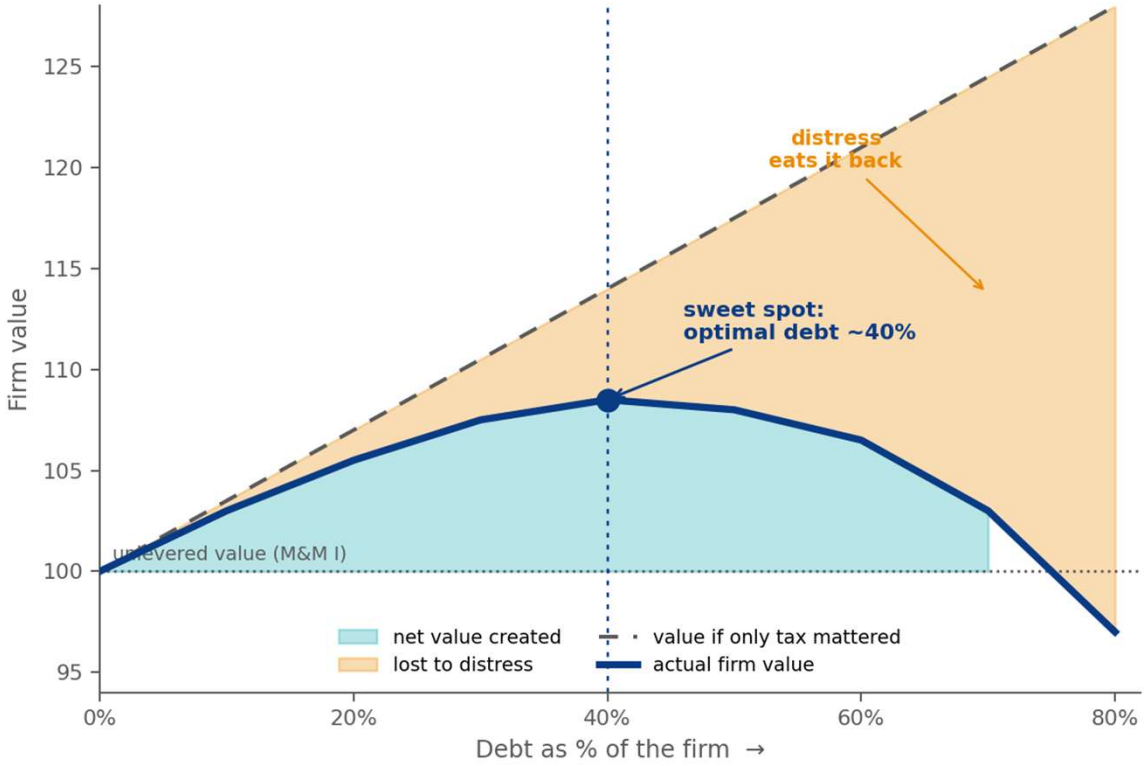
Tax shield (value UP)

Interest is tax-deductible. Borrow £100m at 5% = £5m interest; at 25% tax that saves £1.25m a year.

Distress cost (value DOWN)

Too much debt raises the chance of bankruptcy, and that expected cost eats into value.

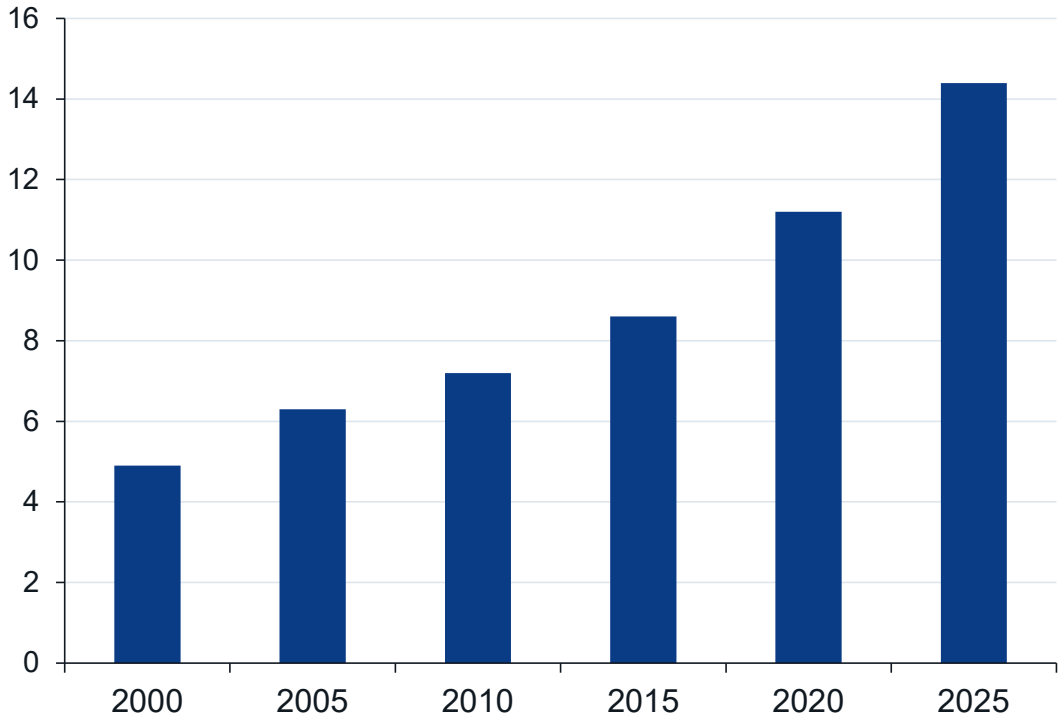
Net them and you get a sweet spot: the debt level that maximises value.



CLICK How is capital structure set in practice? Ask AI ↗

Zoom out: the world is loading up on debt

US company borrowing has roughly tripled since 2000. Why?



US non-financial corporate debt (bonds + loans), $ trillions.

Why is debt growing?

Cheap money: a decade of near-zero rates made borrowing almost free.

The tax shield: interest is tax-deductible, so debt genuinely lowers the tax bill.

Funding buybacks: firms even borrow to buy back their own shares, yes, Apple too.

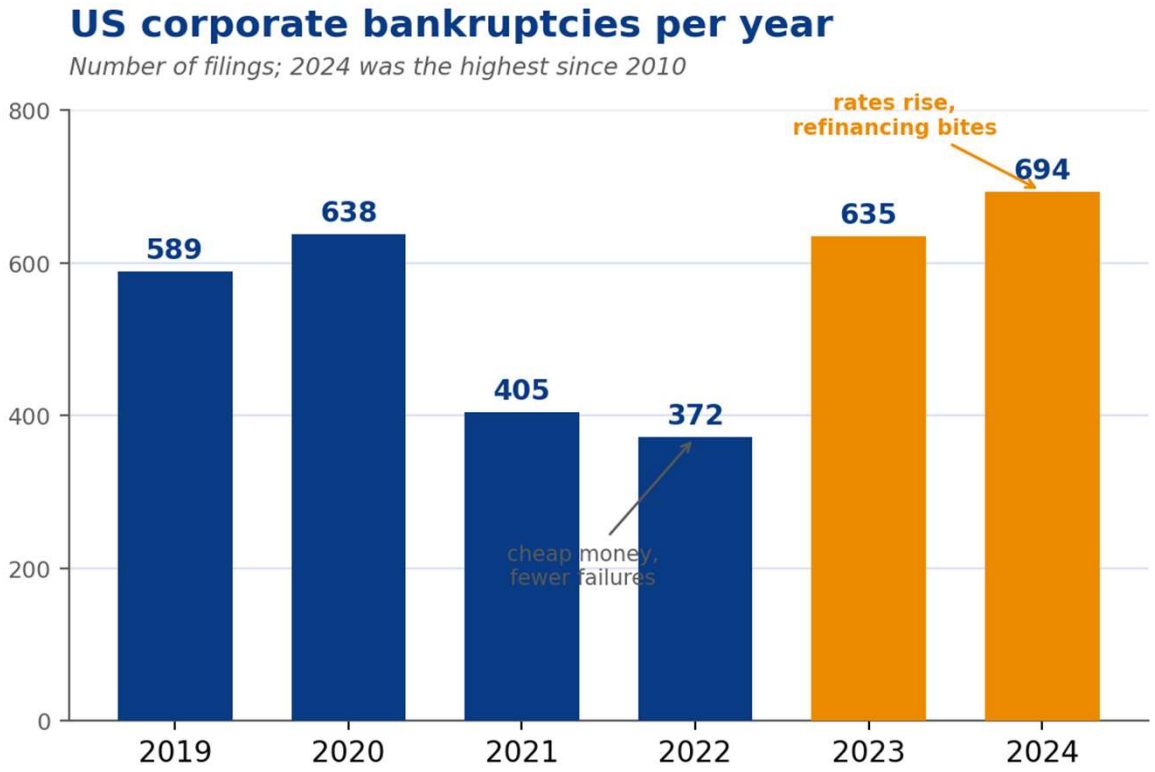
The catch: since 2022, higher rates make all this debt costlier to refinance.

DEEPER READ Companies pile on record debt as global borrowing hits new highs (OECD, 2026) ↗

Source: US Federal Reserve Financial Accounts (non-financial corporate debt).

And now the bill arrives: more firms are failing

Cheap-money debt, met by higher rates, is pushing corporate failures up



Why it matters

Firms borrowed heavily when money was cheap.

When rates jumped, refinancing that debt got far more expensive.

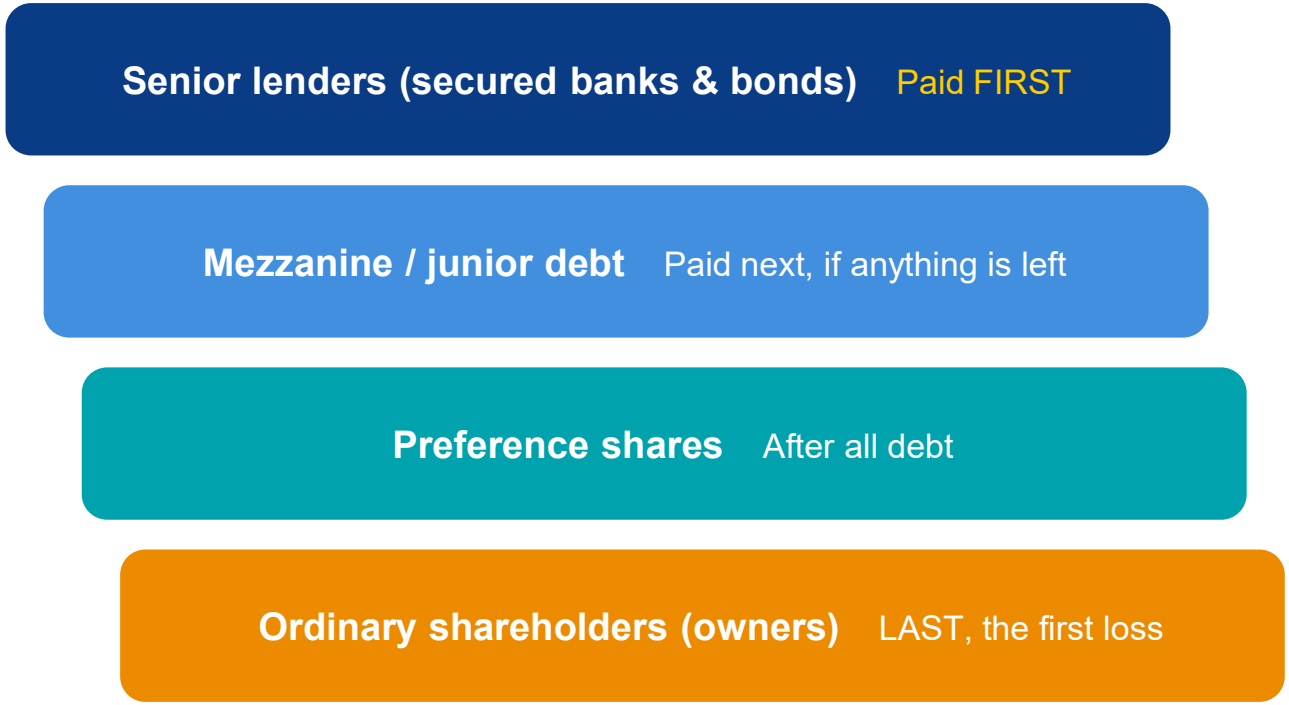
Those who cannot roll it over affordably tip into distress.

The corporate default rate roughly doubled, from about 2% in 2021 to 4-5% by 2024. (S&P Global)

Bottom line: **cheap debt is easy to take on and risky to carry.**

When it goes wrong: the cash-flow waterfall

Who gets paid, and in what order, when a company collapses



Why the middle layers?

Senior debt is cheap but limited; equity is dear. These fill the gap.

Mezzanine / junior debt

Ranks behind senior lenders, so it is riskier and pays a higher rate. It lets a firm borrow more than banks alone allow. Common in buyouts.

Preference shares

A debt-equity hybrid: a fixed dividend, paid before ordinary shares, usually no vote. Raises cash without adding debt or losing control. In 2008 Warren Buffett bought $5bn of Goldman Sachs 'prefs' paying 10% a year.

Bottom line: **Carillion, 2018:** the UK construction giant collapsed owing billions. Shareholders were wiped out completely; even many lenders recovered about a penny per pound. This is why equity is called the first loss.

Carillion entered compulsory liquidation in January 2018 (House of Commons Library; PwC).

A last truth about debt: it has to be refinanced

The real question is not 'can they repay?' but 'can they roll it over?'

Debt must be repaid…

…except perpetuals, bonds that never mature and just pay interest forever.

Most big borrowers, governments and large firms, never really repay. They issue new debt to pay off the old.

So the real risk is refinancing risk: can they roll it over, and at what interest rate?



When rollover fails

If lenders lose confidence, a perfectly sound borrower can still fail to refinance, and fail.
Thames Water, 2024-2025: Britain's biggest water company, with millions of paying customers, nearly sank under about £20bn of debt it could not refinance. It has survived only on emergency creditor lifelines. (Bloomberg, 2025)
It links to last week: when rates rise, rolling over old debt suddenly costs far more.
This is the thread into banks, money and liquidity, weeks 3 and 4.

Bottom line: **A borrower rarely dies because it must repay. It dies because it cannot refinance.**

CLICK What is refinancing risk? Ask AI ↗

So why would anyone buy equity?

Last in the queue, but it owns the whole upside



Limited liability

You can lose what you put in, but no more. Your other assets are safe.



Unlimited upside

Debt's return is capped at its interest. Equity owns all the growth, forever.



Cash returns

Owners share the profits through dividends and share buybacks.

Real numbers: Apple has bought back over $700bn of its own shares in a decade, and approved another ~$100bn in 2025; Shell and BP return billions yearly. Dividends and buybacks are how the upside reaches owners, the reward for standing last in the queue.

DEEPER READ S&P 500 buybacks on track for a record $1 trillion in 2025 (S&P Dow Jones Indices) ↗

So how does the money reach the company?

It travels through middlemen, and you read them by their balance sheet too

Agency (a broker)

Introduces the two sides; the asset passes straight between them.

The deal never lands on ITS OWN balance sheet, so it takes no risk on it.

Earns a fee or commission.

A stockbroker (Hargreaves Lansdown), a corporate-finance adviser, a P2P platform. Like an estate agent: sells the house, never owns it.

Principal (transforming)

Stands in the middle as counterparty, taking the asset onto its OWN balance sheet and transforming it.

Bank: turns your deposit (a liability) into a loan (an asset).

Market maker: buys the share onto its own books, then resells it, keeping the bid-ask spread.

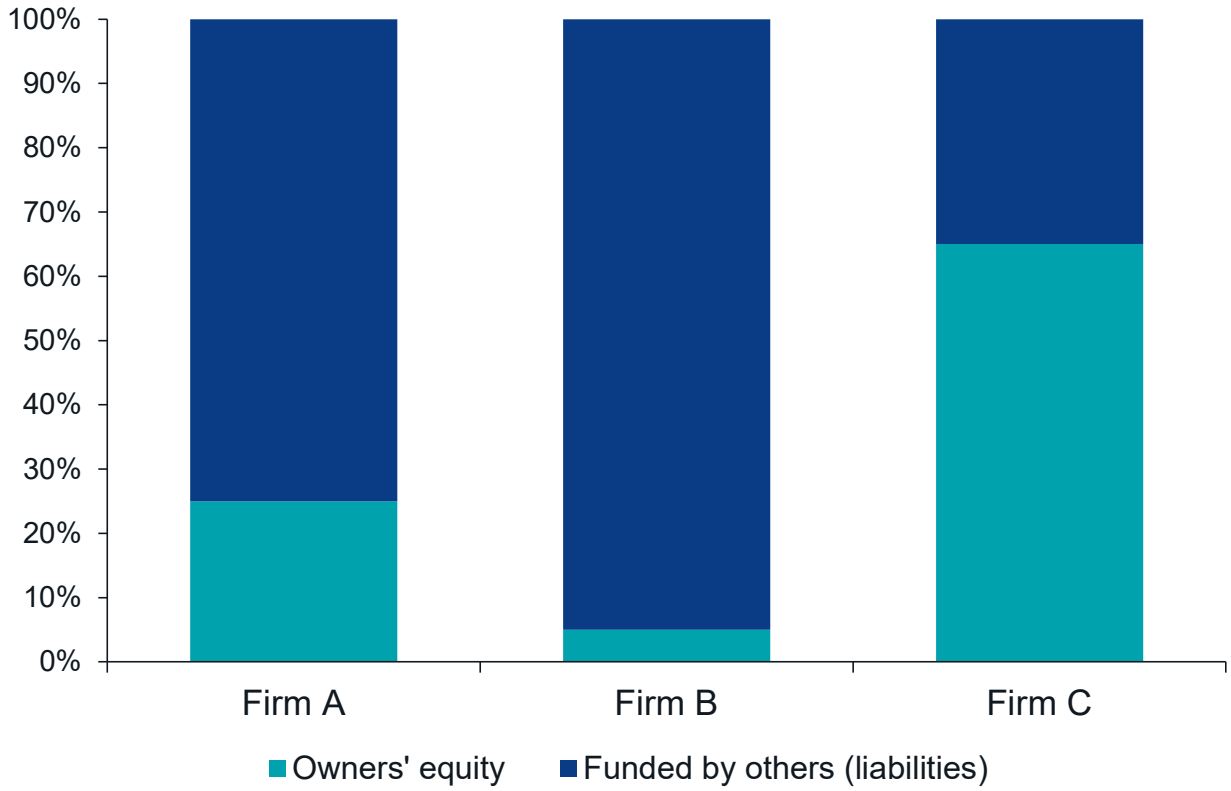
Insurer: takes your premium and your risk; the promise to pay is its liability, the premiums are invested as its assets.

Each one bears the risk, and earns the spread rather than a simple fee.

Bottom line: The test is not 'does it have a balance sheet' (every firm does). It's: does the DEAL sit on its own balance sheet? Broker no, transformer yes.

Balance-sheet X-ray: which one is the bank?

Three real firms, funding shown as a share of total. Vote A, B or C.



Join at vevox.app

The reveal: banks run on borrowed money

Same three firms, now named

Firm A = Tesco

A supermarket

~25% equity

Stores and stock, funded with a moderate mix of debt and equity.

Firm B = Lloyds

A bank

~5% equity

Almost all of it is depositors' and lenders' money. A wafer-thin cushion.

Firm C = Schroders

An asset manager

~65% equity

Light, safe balance sheet; it manages clients' money off its own books.

Same accounting, three different funding stories. Tesco's other ~75% is mainly suppliers it has not paid yet, plus some loans and bonds. Lloyds' ~95% is customer deposits and money borrowed from other lenders: for a bank, the debt IS the business. Schroders barely borrows at all; its small non-equity is just everyday operating bills, and the clients' billions it invests sit off its own books, so they never become its debt. That wafer-thin ~5% cushion is what makes the bank the fragile one, and why a bank can collapse. That is next week's whole story.

Stylised from 2024 annual reports (Tesco, Lloyds Banking Group, Schroders).

Quick poll, grab your phones

You buy a Tesco share on your phone. Who gets your money?

- **A** Tesco, the company
- **B** The investor who sold you the share
- **C** The government
- **D** The Bank of England

Join at vevox.app

Where balance sheets meet: the markets

Two jobs: raising new money (primary), and trading what already exists (secondary)

Primary market

Brand-new securities are sold to raise fresh money. This funds real investment.
IPOs, new bond issues, rights issues, underwriting.
New in 2025, worldwide: about $30tn of bonds and $0.6tn of shares issued.
'Funding liquidity': cash reaches the company.

Secondary market

Existing securities change hands between investors. No new money for the company.
Everyday trading on the LSE and NYSE; market makers quote the prices.
Already out there: about $161tn of bonds and $158tn of shares.
'Market liquidity': you can sell whenever you want.

Link back: every bond and share on the balance sheets we drew was issued once in the primary market, and has traded in the secondary market ever since. A firm's debt-versus-equity choice, our Modigliani-Miller question, is decided the moment it issues here. (Market sizes: SIFMA, 2025.)

SpaceX, 12 June 2026: the largest IPO in history. It sold new shares to raise about $75bn at a valuation above $2 trillion, and the stock jumped about 19% on day one. That $75bn is the primary market at work; every trade since is the secondary market.

DEEPER READ [SpaceX makes history as the largest IPO ever; stock gains 19% on debut (NPR, 2026) ↗]

The highlights, know these cold

1. A balance sheet is just: assets = liabilities + equity. It maps any company.
2. The whole system is linked balance sheets: every asset is someone else's liability.
3. Two ways to fund anything: debt (a fixed promise) and equity (ownership, the upside).
4. Leverage magnifies returns AND losses; business risk is inherited, financial risk is chosen.
5. Capital structure: M&M, irrelevant in a perfect world; the tax shield lifts value, distress caps it, so there's an optimum.
6. The waterfall: lenders paid first, equity is the first loss (Carillion).
7. Intermediaries: a broker's deal passes through; a transformer takes it onto its OWN balance sheet. Banks run on ~5% equity.
8. Markets: primary raises new money (IPOs), secondary trades it (liquidity & price discovery).

Your written exam is built from this slide.

Exam check, can you answer these?

Same style as your exam

A firm's assets are £100m and its liabilities are £80m. Its equity is…

A) £180m B) £20m C) £100m

Answer: B. Equity = assets - liabilities = 100 - 80 = £20m.

A company funds itself with more debt. In a BAD year, the owners' return will be…

A) higher B) lower / more negative C) unchanged

Answer: B. Leverage magnifies losses; interest is still owed.

A company collapses. Who is paid LAST?

A) senior lenders B) bondholders C) ordinary shareholders

Answer: C. Equity is the first loss, paid only after everyone else.

Where we are, and where we're going

Lecture 2 of 5

1	**The big picture & the three pillars**	✓ done
2	**Balance sheets & financial intermediaries**	✓ today
3	**Commercial, investment & central banks**	→ next week
4	Money & liquidity	
5	Capital flows & the four sectors	

Today you learned to read any balance sheet. **Next: the one balance sheet built to be fragile, the bank, and how it can bring down an economy.**

Over to you

Before next week

- Read Chapters 2 & 3 (pp. 28-52).
- Look ahead: Chapters 4 & 7.
- Take the Blackboard quiz.
- Read one finance story in the FT this week.

Your Workspace task

1. Pick a bank (Lloyds) and a non-financial firm (Tesco).
2. In LSEG Workspace: search it, open Financials, then Balance Sheet.
3. Note total assets & total equity; work out equity ÷ assets.

Which one runs on the thinner cushion? The gap will surprise you.

YOUR SEMINAR THIS WEEK **Seminar 2: Balance Sheets & Intermediation**

It goes deeper on today: real company balance sheets, agency vs transforming intermediaries, capital structure, and the products and markets.

To prepare: read Chapters 2 & 3.